

## Exercise Sheet 5

### Propositional Logic – Semantics & Equivalences

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Note that question 2 is marked as being assessed.

1. Provide an example of:
  - a satisfiable formula, which is not valid, containing at least a conjunction, a disjunction, a negation, and an implication
  - a falsifiable formula, which is not unsatisfiable, containing at least a conjunction, a disjunction, a negation, and an implication
  - an unsatisfiable formula, containing at least a conjunction, a disjunction, a negation, and an implication
  - a valid formula, containing at least a conjunction, a disjunction, a negation, and an implication
2. **assessed:** Check whether the following formula is valid using a truth table:  $(q \vee r \rightarrow p) \wedge (q \rightarrow r) \wedge \neg r$
3. Convert  $(q \vee r \rightarrow p) \wedge (q \rightarrow r) \wedge \neg r$  to a DNF using a truth table.
4. Convert  $(q \vee r \rightarrow p) \wedge (q \rightarrow r) \wedge \neg r$  to a CNF using a truth table.
5. Convert  $(q \vee r \rightarrow p) \wedge (q \rightarrow r) \wedge \neg r$  to a CNF using standard equivalences (i.e., using the equivalences listed on slide 10 of lecture 8), as we did on slide 26 of lecture 8.
6. Check whether the following formula is valid using a truth table:  $(q \vee r \rightarrow p) \vee (q \rightarrow r) \vee \neg r$
7. Consider the following solution to the simple Sudoku puzzle discussed in exercise sheet 2:

$$\begin{aligned}
 & (p_0 \vee p_1) \wedge (q_0 \vee q_1) \wedge (r_0 \vee r_1) \wedge (s_0 \vee s_1) \\
 & \wedge (\neg p_0 \vee \neg p_1) \wedge (\neg q_0 \vee \neg q_1) \wedge (\neg r_0 \vee \neg r_1) \wedge (\neg s_0 \vee \neg s_1) \\
 & \wedge (p_0 \vee q_0) \wedge (r_0 \vee s_0) \wedge (p_1 \vee q_1) \wedge (r_1 \vee s_1) \wedge (p_0 \vee r_0) \wedge (q_0 \vee s_0) \wedge (p_1 \vee r_1) \wedge (q_1 \vee s_1)
 \end{aligned}$$

Find a valuation that satisfies this formula, and explain why it does satisfy the formula. Do not draw a truth table. How many rows would the truth table have if you had to draw it?